SF2943 Part A — Method and Mathematics Companion

Electricity load forecasting for Swedish bidding zone SE_3

Purpose. This is a reading aid for `project_part_a.ipynb`. It collects the mathematics used in the pipeline (clean → fit → forecast) with citations to Brockwell & Davis (*Introduction to Time Series and Forecasting*, 2nd ed.), the course text. It does *not* reproduce numerical results or plots — those live in the notebook.

1 Stationarity and the decomposition model

Weak stationarity [B&D §1.4]

A time series $\{X_t\}_{t\in\mathbb{Z}}$ is *weakly stationary* if the mean function is constant and the covariance depends only on the lag:

$$\mu_X(t) = \mathbb{E}[X_t] = \mu, \qquad \gamma_X(t, t+h) = \mathrm{Cov}(X_t, X_{t+h}) = \gamma_X(h). \tag{1}$$

The *autocorrelation function (ACF)* is $\rho_X(h) = \gamma_X(h)/\gamma_X(0)$. Stationarity is what makes the "one realization, many lags" trick work: sample averages along the time axis estimate population quantities. Raw electricity load satisfies neither condition — it has a secular trend, a strong winter/summer cycle, a weekday/weekend cycle, and amplitude that grows with the level — so the cleaning stage is devoted to transforming it into something $\hat{Y}_t$ for which stationarity is a defensible assumption.

Classical decomposition [B&D §1.5]

The course model writes

$$X_t = m_t + s_t + Y_t \tag{2}$$

where m_t is a deterministic trend, s_t is a deterministic seasonal component of known period d satisfying $s_{t-d} = s_t$ and $\sum_{j=1}^{d} s_j = 0$ (zero-mean over one period, by convention), and Y_t is a zero-mean residual. Electricity load has two periods, so we generalise to

$$X_t = m_t + s_t^{(\text{week})} + s_t^{(\text{year})} + Y_t, \qquad d_{\text{week}} = 7,\ d_{\text{year}} = 365. \tag{3}$$

The cleaning procedure estimates $\hat{m}_t$, $\hat{s}_t^{(\text{week})}$, $\hat{s}_t^{(\text{year})}$ and returns $\hat{Y}_t = X_t - \hat{m}_t - \hat{s}_t^{(\text{week})} - \hat{s}_t^{(\text{year})}$.

Variance stabilisation: Box–Cox with $\lambda = 0$ [B&D §1.5]

The amplitude of daily load fluctuations grows with the level (winters are bigger in absolute MW than summers). The Box–Cox transform with parameter $\lambda = 0$ is the natural log,

$$X_t \longmapsto \log X_t, \tag{4}$$

which turns multiplicative seasonality into additive seasonality — i.e. a form that matches (**??**) exactly. Applied *first*, before decomposition.

Harmonic regression for $\hat{s}_t$ [B&D §1.3]

For period d we model the seasonal component as a truncated Fourier series,

$$s_t \;=\; a_0 + \sum_{j=1}^{k} \Big[a_j \cos(\lambda_j t) + b_j \sin(\lambda_j t)\Big], \qquad \lambda_j \;=\; \frac{2\pi j}{d}, \tag{5}$$

with k the number of harmonics retained. The coefficients $\{a_j, b_j\}$ are estimated by ordinary least squares: we form the design matrix

$$\mathbf{Z} \;=\; \big[\mathbf{1},\ \cos(\lambda_1 t),\ \sin(\lambda_1 t),\ \ldots,\ \cos(\lambda_k t),\ \sin(\lambda_k t)\big] \tag{6}$$

and compute $\hat{\boldsymbol{\beta}} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{y}$. This is a linear projection onto a finite-dimensional basis of trigonometric polynomials — the cleanest possible setting for least squares.

Choices for this project. Yearly: $d = 365$, $k = 2$ (fundamental + first harmonic, enough to capture the asymmetric winter peak without over-fitting). Weekly: $d = 7$, $k = 3$ (three harmonics are enough resolution for the near-rectangular weekday/weekend block shape). The estimated coefficients are printed in the notebook and copied into the summary file.

Polynomial trend [B&D §1.5]

After removing the seasonal components we fit the remaining slowly-varying level by a low-order polynomial:

$$\hat{m}_t \;=\; \beta_0 + \beta_1 t + \beta_2 t^2 + \cdots, \tag{7}$$

also by ordinary least squares. Default degree: 1 (linear). The notebook escalates to degree 2 (quadratic) only if the linear residuals show visible curvature — concretely, if the mean of any third of the residual series exceeds 10% of the residual standard deviation. This heuristic is a one-line stand-in for the visual check B&D describes.

Differencing: an alternative route [B&D §1.5.2]

With the backshift operator $BX_t = X_{t-1}$, define the lag-d difference

$$\nabla_d X_t \;=\; X_t - X_{t-d} \;=\; (1 - B^d)X_t. \tag{8}$$

Applied to the decomposition (**??**) with s_t of period d,

$$\nabla_d X_t = (m_t - m_{t-d}) + (Y_t - Y_{t-d}),$$

so the seasonal component is killed exactly. A linear trend is killed by $\nabla_1 = 1 - B$, a quadratic trend by ∇_1^2. The project uses classical decomposition as the primary route (it yields explicit $\hat{m}_t$ and $\hat{s}_t$ to report) and differencing as a sanity check: the residual ACFs from $\hat{Y}_t$ and from $\nabla_1\nabla_7 \log X_t$ should tell the same story.

Sample ACF and the $\pm 1.96/\sqrt{n}$ bounds [B&D §1.4.1]

The sample autocovariance is

$$\hat{\gamma}(h) \;=\; \frac{1}{n}\sum_{t=1}^{n-|h|}(x_{t+|h|} - \bar{x})(x_t - \bar{x}), \tag{9}$$

and $\hat{\rho}(h) = \hat{\gamma}(h)/\hat{\gamma}(0)$ is the sample ACF. The divisor n (rather than $n - h$) is used to keep the sample covariance matrix nonnegative definite — this matters when we later feed $\hat{\gamma}$ into Yule–Walker. For an iid sequence with finite variance,

$$\hat{\rho}(h) \;\overset{\text{approx}}{\sim}\; \mathcal{N}\big(0,\ \tfrac{1}{n}\big), \qquad h \neq 0, \tag{10}$$

so approximately 95% of the $\hat{\rho}(h)$ lie inside $\pm 1.96/\sqrt{n}$. These are the blue dashed lines drawn on every ACF plot in the notebook. A correctly cleaned residual series has a sample ACF that stays mostly inside these bounds, with no slow decay (residual trend) and no spikes at period multiples (residual seasonality).

Ljung–Box portmanteau test [B&D §1.6]

The joint statistic

$$Q_{\mathrm{LB}} \;=\; n(n+2)\sum_{j=1}^{h}\frac{\hat{\rho}^2(j)}{n-j} \tag{11}$$

replaces eyeballing individual $\hat{\rho}(h)$. Under the null that $\{X_t\}$ is iid, $Q_{\mathrm{LB}} \overset{\text{approx}}{\sim} \chi^2(h)$. We report it at $h = 20$ and $h = 40$. For the *cleaning* residuals we *expect* iid to be rejected — that is what justifies fitting an ARMA model in the next stage. For the *ARMA* residuals (Section **??** below) we apply a degrees-of-freedom correction, as discussed there.

2 ARMA identification and estimation

ARMA(p, q) and causality/invertibility [B&D §3.1]

A causal ARMA(p, q) process satisfies

$$\phi(B)X_t \;=\; \theta(B)Z_t, \qquad \{Z_t\} \sim \mathrm{WN}(0,\sigma^2), \tag{12}$$

where $\phi(z) = 1-\phi_1 z-\cdots-\phi_p z^p$ and $\theta(z) = 1+\theta_1 z+\cdots+\theta_q z^q$. The process is *causal* if every root of $\phi(z)$ satisfies $|z| > 1$ (equivalently: X_t is an absolutely convergent one-sided moving average of the innovations $\{Z_s : s \le t\}$), and *invertible* if every root of $\theta(z)$ satisfies $|z| > 1$ (equivalently: Z_t can be written as an absolutely convergent one-sided function of $\{X_s : s \le t\}$). Both conditions must hold for the estimation and prediction machinery to be well-defined; the notebook checks them on every fitted candidate by computing the roots of $\hat{\phi}$ and $\hat{\theta}$.

Identification from sample ACF and PACF [B&D §3.2.3, 6.2]

The shapes of $\hat{\rho}$ and of the sample partial autocorrelation $\hat{\alpha}$ diagnose the order:

Process	Sample ACF	Sample PACF
AR(p)	tails off	**cuts off after lag** p
MA(q)	**cuts off after lag** q	tails off
ARMA(p,q)	tails off	tails off

"Cuts off" means: drops inside the $\pm 1.96/\sqrt{n}$ bounds and stays there. This is the *first-pass* tool for picking (p,q) candidates before any numerical optimisation is run. The notebook uses it to select 3–5 candidate models, which are then compared by AICC.

Yule–Walker estimation for AR(p) [B&D §5.1.1]

Multiplying the AR equation $X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t$ by X_{t-j} and taking expectations gives, for $j \ge 1$,

$$\gamma(j) = \phi_1\gamma(j-1) + \cdots + \phi_p\gamma(j-p),$$

and evaluating at $j = 1,\ldots,p$ yields the Yule–Walker system

$$\Gamma_p\boldsymbol{\phi} \;=\; \boldsymbol{\gamma}_p, \qquad \sigma^2 \;=\; \gamma(0) - \boldsymbol{\phi}^\top\boldsymbol{\gamma}_p, \tag{13}$$

where $\Gamma_p = [\gamma(i-j)]_{i,j=1}^{p}$, $\boldsymbol{\gamma}_p = (\gamma(1),\ldots,\gamma(p))^\top$, and $\boldsymbol{\phi} = (\phi_1,\ldots,\phi_p)^\top$. The sample version replaces γ with $\hat{\gamma}$:

$$\hat{\boldsymbol{\phi}} \;=\; \hat{\Gamma}_p^{-1}\hat{\boldsymbol{\gamma}}_p, \qquad \hat{\sigma}^2 \;=\; \hat{\gamma}(0) - \hat{\boldsymbol{\phi}}^\top\hat{\boldsymbol{\gamma}}_p. \tag{14}$$

Two properties are useful: (i) $\hat{\Gamma}_p$ is automatically invertible because $\hat{\gamma}$ is built with divisor n, guaranteeing nonnegative definiteness; (ii) $\hat{\boldsymbol{\phi}}$ is asymptotically equivalent to MLE for AR processes, so Yule–Walker is used in the notebook as a cross-check on the MLE for the best AR candidate, not as the primary estimator.

Gaussian MLE via the innovations algorithm [B&D §5.2]

For models with $q > 0$, Yule–Walker is not enough and we use Gaussian MLE. The trick that makes the likelihood computable is the *innovations algorithm*: letting $\hat{X}_j$ denote the one-step predictor of X_j from $X_1, \ldots, X_{j-1}$, and $v_{j-1} = \mathbb{E}[(X_j - \hat{X}_j)^2]$ its mean squared error, the *innovations* $U_j = X_j - \hat{X}_j$ are uncorrelated by construction, so the joint Gaussian density factorises:

$$L(\phi, \theta, \sigma^2) \;=\; (2\pi)^{-n/2} \prod_{j=1}^{n} v_{j-1}^{-1/2} \exp\left\{ -\frac{1}{2} \sum_{j=1}^{n} \frac{U_j^2}{v_{j-1}} \right\}. \tag{15}$$

Maximising this numerically over (ϕ, θ, σ^2) is the Gaussian MLE; `statsmodels.tsa.arima.model.ARIMA` does exactly this under the hood. The notebook comments every ARMA fit with a reference to this section.

Order selection: AICC, not AIC [B&D §5.5.2]

Among candidate models we pick the one minimising

$$\mathrm{AICC} \;=\; -2 \ln L(\hat{\phi}, \hat{\theta}, \hat{\sigma}^2) + \frac{2(p+q+1)\, n}{n-p-q-2}, \tag{16}$$

where n is the sample size and the "+1" on $p+q+1$ counts σ^2. Compared to the uncorrected AIC, AICC adds the finite-sample bias term

$$\frac{2k(k+1)}{n-k-1}, \qquad k = p+q+1, \tag{17}$$

which disappears as $n \to \infty$ but meaningfully penalises overparameterised models in finite samples. This is the only information criterion the course uses: AIC underpenalises overfitting in finite samples, and AICC is the standard fix. The notebook computes it as $\mathrm{AICC} = \mathrm{AIC} + 2k(k+1)/(n-k-1)$ for each candidate.

Ljung–Box on ARMA residuals [B&D §5.3]

After a model is fit, we recompute the portmanteau statistic on its residuals $\hat{R}_t$ and compare against a χ^2 distribution with *corrected* degrees of freedom:

$$Q_{\mathrm{LB}}(\hat{R}) \;\overset{\text{approx}}{\sim}\; \chi^2\big(h-p-q\big). \tag{18}$$

The $p+q$ degrees of freedom are "spent" on estimating the ARMA parameters; ignoring this correction leads to spuriously conservative p-values. The notebook passes `model_df = p + q` to `statsmodels.stats.diagnostic.acorr_ljungbox`. If the residual test rejects, we either accept the failure and discuss what structure is left, or we go back and try a different (p, q) — we do not paper over it.

3 Linear prediction and forecasting

Best linear predictor [B&D §2.5]

For a zero-mean stationary process and given $X_1, \ldots, X_n$, the best linear (i.e. minimum mean squared error among linear) predictor of X_{n+h} is

$$P_n X_{n+h} \;=\; \boldsymbol{\phi}_n^\top \mathbf{X}_n \;=\; \phi_{n1} X_n + \cdots + \phi_{nn} X_1, \tag{19}$$

where the coefficients satisfy the prediction equations

$$\Gamma_n \boldsymbol{\phi}_n \;=\; \boldsymbol{\gamma}_n(h), \qquad \boldsymbol{\gamma}_n(h) = (\gamma(h), \gamma(h+1), \ldots, \gamma(h+n-1))^\top, \tag{20}$$

with MSE

$$v_n(h) \;=\; \mathbb{E}\big[(X_{n+h} - P_n X_{n+h})^2\big] \;=\; \gamma(0) - \boldsymbol{\phi}_n^\top \boldsymbol{\gamma}_n(h). \tag{21}$$

For non-zero-mean series, predict $X_t - \mu$ and add μ back.

Innovations algorithm for prediction [B&D §2.5.4]

Rewriting the predictor in terms of innovations $U_j = X_j - \hat{X}_j$ instead of raw observations,

$$\hat{X}_{n+1} \;=\; \sum_{j=1}^{n} \theta_{nj} \left(X_{n+1-j} - \hat{X}_{n+1-j} \right), \tag{22}$$

yields a clean recursion because the innovations are uncorrelated by construction — unlike the X_j themselves, which are highly correlated. For MA and ARMA processes this is the natural route; for pure AR the Durbin–Levinson recursion [B&D §2.5.3] does the same job directly on the observations.

h-step ARMA forecast [B&D §3.3.1]

For $n > \max(p, q)$ and $h \geq 1$,

$$P_n X_{n+h} \;=\; \sum_{i=1}^{p} \phi_i \, P_n X_{n+h-i} \;+\; \sum_{j=h}^{q} \theta_{n+h-1,\,j} \left(X_{n+h-j} - \hat{X}_{n+h-j} \right). \tag{23}$$

The second sum vanishes once $h > q$, so long-horizon forecasts follow the AR recursion alone and decay geometrically towards the process mean. The prediction MSE $v_n(h)$ satisfies

$$v_n(h) \;\longrightarrow\; \mathrm{Var}(X_t) \quad \text{as } h \to \infty,$$

reflecting the fact that the forecast carries no information about the distant future beyond the unconditional mean. Visually, this is the characteristic "point forecast flattens, band widens" shape on the forecast plot in the notebook.

Prediction intervals

Under Gaussian innovations, an approximate 95% prediction interval is

$$P_n X_{n+h} \pm 1.96 \sqrt{v_n(h)}. \tag{24}$$

In the notebook this interval is computed on the *residual* (log-MW) scale and then transformed to MW by adding the extrapolated seasonal and trend components and exponentiating. Because exp is nonlinear, the MW interval is asymmetric (log-normal) — this is noted as an approximation but used as-is, since the course does not introduce bias-corrected log-normal intervals.

4 The pipeline in one page

Bringing it all together, the notebook executes:

1. X_t (raw hourly MW) $\to$ daily mean (preserves MW units);
2. $\to \log X_t$ (Box–Cox, $\lambda = 0$, variance stabilisation);
3. $\to$ subtract harmonic yearly seasonal $\hat{s}_t^{(\text{year})}$ ($d = 365$, $k = 2$, OLS on (**??**));
4. $\to$ subtract harmonic weekly seasonal $\hat{s}_t^{(\text{week})}$ ($d = 7$, $k = 3$);
5. $\to$ subtract polynomial trend $\hat{m}_t$ (OLS, degree 1 or 2 by curvature check);
6. $\to$ residual series $\hat{Y}_t$ (target: stationary);
7. $\to$ diagnose via sample ACF, PACF, Ljung–Box [B&D §1.6];

8. $\rightarrow$ identify candidate ARMA(p, q) from ACF/PACF shape [B&D §3.2.3];
9. $\rightarrow$ fit each candidate by Gaussian MLE [B&D §5.2]; verify causality and invertibility; compute AICC (**??**); pick the minimum; cross-check the best AR candidate against Yule–Walker (**??**);
10. $\rightarrow$ diagnose ARMA residuals via Ljung–Box with $\mathrm{df} = h - p - q$ [B&D §5.3];
11. $\rightarrow$ refit chosen ARMA on the training window; compute h-step forecast (**??**) for $h = 1, \ldots, 30$ and its $v_n(h)$-based 95% band;
12. $\rightarrow$ invert the cleaning transforms (add trend + seasonals, then exp) to recover MW;
13. $\rightarrow$ report RMSE, MAE, and empirical coverage of the held-out 30-day test window.

Each step in the notebook is commented with the corresponding B&D section, so the notebook can be read as an annotated implementation of the theory above.